Lakshya JEE 2.0 (2024)

Limits

Assignment

1. The value of $\lim_{x\to 1}\frac{\sqrt[13]{x}-\sqrt[7]{x}}{\sqrt[5]{x}-\sqrt[3]{x}}$ is

(1) $\frac{9}{13}$ (2) $\frac{5}{91}$

(3) $\frac{9}{91}$ (4) $\frac{45}{91}$

2. The value of $\lim_{x\to 1}\frac{\left[\sum_{K=1}^{100}x^k\right]-100}{x-1}$ is

(1) 4950 (2) 5050

(3) 5051 (4) 5151

3. The value of $\lim_{x\to 1}\left(\frac{p}{1-x^p}-\frac{q}{1-x^q}\right)p,q\in N,$ is

(1) $\frac{p-q}{2}$ (2) $\frac{p+q}{2}$

(3) $\frac{q-p}{2}$ (4) None of these

4. The value of $\lim_{x\to\frac{3\pi}{4}}\frac{1+\sqrt[3]{\tan x}}{1-2\cos^2 x}$ is

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $-\frac{1}{2}$ (4) $-\frac{1}{3}$

5. The value of $\lim_{x\to 0}(\ln(1+\sin^2 x))\cot(\ln^2(1+x))$ is

(1) 0 (2) 2

(3) 1 (4) $\frac{1}{2}$

6. The value of $\lim_{x\to\infty}\frac{2\sqrt{x}+3x^{1/3}+5x^{1/5}}{\sqrt{3x-2}+(2x-3)^{1/3}}$ is

(1) 1 (2) $\frac{1}{\sqrt{3}}$

(3) $\frac{2}{\sqrt{3}}$ (4) $\sqrt{3}$

7. The value of $\lim_{x\to 0}\frac{\sec 4x-\sec 2x}{\sec 3x-\sec x}$ is

(1) 1 (2) $\frac{2}{3}$

(3) $\frac{3}{2}$ (4) 2

8. The sum of an infinite geometric series whose first term is the limit of the function $f(x)=\frac{\tan x-\sin x}{\sin^3 x}$ as $x\to 0$ and whose common ratio is the limit of the function $g(x)=\frac{1-\sqrt{x}}{(\cos^{-1}x)^2}$ as $x\to 1$, is

(1) $\frac{1}{3}$ (2) $\frac{1}{4}$

(3) $\frac{1}{2}$ (4) $\frac{2}{3}$

9. The value of $\lim_{x\to\infty}\left(x-\ln\left(\frac{e^x+e^{-x}}{2}\right)\right)$ is

(1) 0 (2) ln 2

(3) 1 (4) 2 ln 2

10. The value of

$\lim_{x\to 0}\frac{8}{x^8}\left[1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right]$ is

(1) $\frac{1}{16}$ (2) $\frac{1}{32}$

(3) $\frac{1}{64}$ (4) $\frac{1}{8}$

11. The value of $\lim_{\theta\to\frac{\pi}{4}}\frac{\sqrt{2}-\cos\theta-\sin\theta}{(4\theta-\pi)^2}$ is

(1) $\frac{1}{2\sqrt{2}}$ (2) $\frac{1}{4\sqrt{2}}$

(3) $\frac{1}{8\sqrt{2}}$ (4) $\frac{1}{16\sqrt{2}}$

12. The value of $\lim_{x\to\frac{\pi}{2}} \dfrac{2^{-\cos x}-1}{x(x-\frac{\pi}{2})}$ is

(1) $\dfrac{\ln 2}{\pi}$ (2) $\dfrac{2\ln 2}{\pi}$

(3) $\dfrac{\ln 8}{\pi}$ (4) $\dfrac{\ln 16}{\pi}$

13. The value of $\lim_{x\to 1} \dfrac{(\ln(1+x)-\ln 2)(3.4^{x-1}-3x)}{[(7+x)^{\frac{1}{3}}-(1+3x)^{\frac{1}{2}}].\sin(x-1)}$ is

(1) $\dfrac{3}{4}\ln\dfrac{4}{e}$ (2) $-\dfrac{3}{4}\ln\dfrac{4}{e}$

(3) $\dfrac{9}{4}\ln\dfrac{4}{e}$ (4) $-\dfrac{9}{4}\ln\dfrac{4}{e}$

14. The value of $\lim_{x\to\infty}\sum_{r=2}^{x}\left((r+1)\sin\dfrac{\pi}{r+1}-r\sin\dfrac{\pi}{r}\right)$ is

(1) $\pi-2$ (2) $\dfrac{\pi}{2}-1$

(3) $2\pi-2$ (4) $\dfrac{\pi}{4}-\dfrac{1}{2}$

15. The value of $\lim_{x\to-\infty} \dfrac{(3x^4+2x^2)\sin\frac{1}{x}+|x|^3+5}{|x|^3+|x|^2+|x|+1}$ is

(1) 2 (2) -2

(3) 1 (4) -1

16. The value of $\lim_{x\to 3} \dfrac{(x^3+27)\ln(x-2)}{x^2-9}$ is

(1) 1 (2) 3

(3) 6 (4) 9

17. The value of $\lim_{x\to 0} \dfrac{27^x-9^x-3^x+1}{\sqrt{2}-\sqrt{1+\cos x}}$ is

(1) $8\sqrt{2}(\ln 3)^2$ (2) $4\sqrt{2}(\ln 3)^2$

(3) $2\sqrt{2}(\ln 3)^2$ (4) $\sqrt{2}(\ln 3)^2$

18. The value of $\lim_{x\to 0}\left[\dfrac{(1+x)^{1/x}}{e}\right]^{1/x}$ is

(1) $\sqrt{e}$ (2) $\dfrac{1}{\sqrt{e}}$

(3) $\dfrac{1}{e}$ (4) e

19. The value of $\lim_{n\to\infty}\left(\dfrac{\sqrt{x^2+x}-1}{x}\right)^{2\sqrt{x^2+x}-1}$ is

(1) $\sqrt{e}$ (2) $\dfrac{1}{\sqrt{e}}$

(3) $\dfrac{1}{e}$ (4) e

20. The value of $\lim_{x\to 1}\left(\tan\dfrac{\pi x}{4}\right)^{\tan\frac{\pi x}{2}}$ is

(1) e (2) e^{-1}

(3) $\sqrt{e}$ (4) $\dfrac{1}{\sqrt{e}}$

21. $\lim_{x\to 0}\left[\tan\left(\dfrac{\pi}{4}+x\right)\right]^{1/x}$ is equal to

(1) e^{-1} (2) e

(3) e^2 (4) $\sqrt{e}$

22. $\lim_{x\to 0}\left(\dfrac{3^x-1}{x}\right)$ equals

(1) log 3 (2) 3 log 3

(3) 2 log 3 (4) None of these

23. $\lim_{x\to 0}\dfrac{\cos(\sin x)-1}{x^2}=$

(1) 1 (2) -1

(3) 1/2 (4) $-1/2$

24. $\lim_{x\to 0}\dfrac{e^{1/x}}{e^{\frac{1}{x}}+1}=$

(1) 0

(2) 1

(3) 2

(4) Does not exist

25. $\lim_{x\to 0}\dfrac{a^{\sin x}-1}{b^{\sin x}-1}=$

(1) $\dfrac{a}{b}$ (2) $\dfrac{b}{a}$

(3) $\dfrac{\log a}{\log b}$ (4) $\dfrac{\log b}{\log a}$

26. $\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)=$

(1) $\frac{1}{2}$ (2) ∞

(3) 1 (4) 0

27. $\lim_{x\to\infty}\left(\frac{x+3}{x+1}\right)^{x+1}=$

(1) e^2 (2) e^3

(3) e (4) e^{-1}

28. $\lim_{x\to 0}(1-ax)^{\frac{1}{x}}=$

(1) e (2) e^{-a}

(3) 1 (4) e^a

29. If $\lim_{x\to 0}\frac{x^n-\sin x^n}{x-\sin^n x}$ is non zero definite, then n must be

(1) 1 (2) 2

(3) 3 (4) None of these

30. The values of a and b such that $\lim_{x\to 0}\frac{x(2+a\cos x)+b\sin x}{x^3}=2,$ are

(1) $a=5, b=-3$ (2) $a=-5, b=3$

(3) $a=-5, b=-3$ (4) $a=5, b=3$

31. $\lim_{h\to 0}\frac{\sqrt{3}\sin\left(\frac{\pi}{6}+x\right)-\cos\left(\frac{\pi}{6}+x\right)}{\sqrt{3}x(\sqrt{3}\cos x-\sin x)}=$

(1) $-\frac{1}{3}$ (2) $-\frac{3}{8}$

(3) $-\sqrt{3}$ (4) $\frac{2}{3}$

32. The value of $\lim_{x\to 0}\frac{x\cos x-\log(1+x)}{x^2}$ is

(1) $\frac{1}{2}$

(2) 0

(3) 1

(4) None of these

33. $\lim_{x\to\frac{\pi}{4}}\left(\frac{1-\tan x}{1-\sqrt{2}\sin x}\right)$ equals

(1) 0 (2) 1

(3) -2 (4) 2

34. $\lim_{x\to 0}\frac{3^x-1}{\sqrt{4+x}-2}$ equals

(1) log 9 (2) log81

(3) log 3 (4) log 27

35. $\lim_{x\to 0}\left(\frac{\sin(2+2x)+\sin(2-2x)-2\sin 2}{x\sin x}\right)$ equals to

(1) 4sin 2 (2) $-$ 4sin 2

(3) 1 (4) $-$2sin 2

36. $\lim_{x\to\frac{\pi}{4}}\frac{\int_{\pi/4}^{x}t^2\,dt}{\cos 2x}$ equals to

(1) ∞ (2) $\frac{\pi^2}{32}$

(3) $-\frac{\pi^2}{32}$ (4) $-\frac{\pi^2}{16}$

37. $\lim_{h\to 0}\frac{(x+x)\sec(x+x)-x\sec x}{\sin x}=$

(1) $\sec x(x\tan x+1)$

(2) $x\tan x+\sec x$

(3) $x\sec x+\tan x$

(4) None of these

38. $\lim_{x\to\frac{\pi}{6}}\frac{\sin\left(\frac{\pi}{3}-2x\right)}{2\cos 2x-1}$ is equal to

(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{3}}$

(3) $\sqrt{3}$ (4) $\frac{2}{\sqrt{3}}$

39. $\lim_{x\to 0}\frac{e^{x^4}-\cos 2x^2}{x^4}$ is equal to

(1) $\frac{3}{2}$ (2) $\frac{1}{2}$

(3) $\frac{2}{3}$ (4) -1

40. $\lim_{x\to 0}\frac{x\cos^2 x-\sin^2 x}{x+\sin x}$ equal

(1) $\frac{1}{2}$ (2) $-\frac{1}{2}$

(3) 1 (4) –1

41. Assume that $\lim_{x\to 1} f(\theta)$ exists and $\frac{\theta^2+\theta-2}{\theta+3}\le\frac{f(\theta)}{\theta^2}\le\frac{\theta^2+2\theta-1}{\theta+3}$ holds for certain interval containing the point $\theta=-1$ then $\lim_{x\to -1} f(\theta)$

(1) is equal to $f(-1)$ (2) is equal to 1

(3) is non existent (4) is equal to -1

42. Let $l_1=\lim_{x\to\infty}\sqrt{\frac{x-\cos^2 x}{x+\sin x}}$ and $l_2=\lim_{x\to 0^+}\int_{-1}^{1}\frac{h\,dx}{h^2+x^2}$.

Then

(1) both l_1 and l_2 are less than $\frac{22}{7}$

(2) one of the two limits is rational and other irrational.

(3) $l_2 > l_1$

(4) l_2 is greater than 3 times of l_1

43. $\lim_{x\to c} f(x)$ does not exist when

(1) $f(x)=[[x]]-[2x-1], c=3$

(2) $f(x)=[x]-x,\ c=1$

(3) $f(x)=\{x\}^2-\{-x\}^2,\ c=0$

(4) $f(x)=\frac{\tan(\operatorname{sgn} x)}{\operatorname{sgn} x}, c=0$

Where [x] denotes step up function & {x}fractional part function.

44. Let $f(x)=\begin{cases}\frac{\tan^2\{x\}}{x^2-[x]^2} & \text{for } x>0\\ 1 & \text{for } x=0\\ \sqrt{\{x\}\cot\{x\}} & \text{for } x<0\end{cases}$ where [x] is the step up function and {x} is the fractional part function of x, then :

(1) $\lim_{x\to 0^+} f(x)=1$

(2) $\lim_{x\to 0^-} f(x)=1$

(3) $\cot^{-1}\left(\lim_{x\to 0^-} f(x)\right)^2=1$

(4) f is continuous at x = 1

45. Which of the following limits vanish?

(1) $\lim_{x\to\infty} x^{\frac{1}{4}}\sin\frac{1}{\sqrt{x}}$

(2) $\lim_{x\to\pi/2}(1-\sin x).\tan x$

(3) $\lim_{x\to\infty}\frac{2x^2+3}{x^2+x-5}.\operatorname{sgn}(x)$

(4) $\lim_{x\to 3^+}\frac{[x]^2-9}{x^2-9}$

Where [ ] denotes greatest integer function

46. Let $f(x)=\lim_{x\to\infty}\frac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}}$ then which of the following alternative(s) is/are correct?

(1) $\lim_{x\to\infty} x f(x)=2$

(2) $\lim_{x\to 1} f(x)$ does not exist.

(3) $\lim_{x\to 0} f(x)$ does not exist.

(4) $\lim_{x\to\infty} f(x)$ is equal to zero.

47. If $\lim_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2}-\sqrt{x^4+2x^3-cx^2+3x-d}\right)=4,$ then

(1) a = 2 (2) $b\in R$

(3) c = 5 (4) $d\in R$

48. If $\lim_{x\to 0}\frac{a+b\sin x-\cos x+ce^x}{x^3}$ exists finitely, then

(1) a = 2 (2) b = 1

(3) c = 1 (4) $d\in R$

49. If $f(x)=\begin{cases}\frac{a^{[x]+x}}{[x]+x} & x\neq 0\\ \ln a, & x=0\end{cases}$, then which of the following is correct

(1) $\lim_{x\to 0^-} f(x)=1-\frac{1}{a}$

(2) $\lim_{x\to 0^+} f(x)=\ln a$

(3) $\lim_{x\to 0} f(x)$ does not exist

(4) $f(x)$ is continuous at x = 0

50. Let $f(x)=\left(\frac{x}{2+x}\right)^{2x}$, then

(1) $\lim\limits_{x\to\infty} f(x)=e^{-4}$ (2) $\lim\limits_{x\to 1} f(x)=\frac{1}{9}$

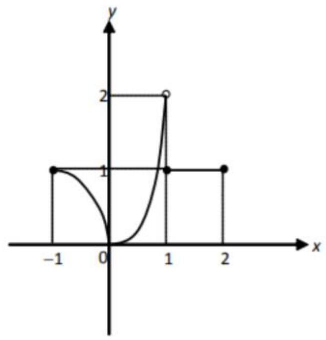
(3) $\lim\limits_{x\to\infty} f(x)=e^{4}$ (4) None of these

51. If $\lim\limits_{x\to 0}\frac{ae^{1/x}+be^{-1/x}}{ce^{1/x}+de^{-1/x}}=2$ then the roots of $bx^2+(a-2c)x-2d=0$, $a, b, c, d\in Q$ are

(1) real (2) integral

(3) rational (4) equal

52. Which of the following statements are true of the function f defined for $-1\le x\le 3$ in the figure shown



(1) $\lim\limits_{x\to -1^+} = 2$

(2) $\lim\limits_{x\to 2} f(x)$ does not exist

(3) $\lim\limits_{x\to 1^-} f(x)=2$

(4) $\lim\limits_{x\to 0^+} f(x)=\lim\limits_{x\to 0^-} f(x)$

53.

Column A	Column B
(A) $\lim\limits_{x\to 0}\frac{1-\cos 2x}{e^{x^2}-e^{x}+x}$ equal	(P) 1
(B) If the value of $\lim\limits_{x\to 0^+}\left(\frac{(3/x)+1}{(3/x)-1}\right)^{1/x}$ can be expressed as $e^{a/b}$ then a + b is	(Q) 2
(C) $\lim\limits_{x\to 0}\frac{\tan^3 x-\tan x^3}{x^5}$ equals	(R) 4
(D) $\lim\limits_{x\to 0}\frac{x+2\sin x}{\sqrt{x^2+2\sin x+1}-\sqrt{\sin^2 x-x+1}}$	(S) 5

54.

Column A	Column B
(A) $\lim\limits_{x\to\infty}\cos^2\left(\pi\left(\sqrt[3]{n^3+n^2+2n}\right)\right)$ where n is an integer, equals	(P) $\frac{1}{2}$
(B) $\lim\limits_{x\to\infty} n\sin\left(2\pi\sqrt{1+n^2}\right)(n\in N)$ equals	(Q) $\frac{1}{4}$
(C) $\lim\limits_{x\to\infty}(-1)^n\sin\left(\pi\sqrt{n^2+0.5n+1}\right)\left(\sin\frac{(n+1)\pi}{4n}\right)$ is (where $n\in N$)	(R) π
(D) If $\lim\limits_{x\to\infty}\left(\frac{x+a}{x-a}\right)^x=e$ where 'a' is some real constant then the value of 'a' is equal to	(S) non existent

55.

Column A	Column B
(A) $\lim\limits_{x\to\infty}\left(\sqrt{x+\sqrt{x}}-\sqrt{x-\sqrt{x}}\right)$ equals	(P) -2
(B) The value of the limit, $\lim\limits_{x\to 0}\frac{\sin 2x-2\tan x}{\ln(1+x^3)}$ is	(Q) -1
(C) $\lim\limits_{x\to 0+}\left(\ln\sin^3 x-\ln(x^4+ex^3)\right)$ equals	(R) 0
(D) Let $\tan(2\pi\lvert\sin\theta\rvert)=\cot(2\pi\lvert\cos\theta\rvert)$, Where $\theta\in R$ and $f(x)=(\lvert\sin\theta\rvert+\lvert\cos\theta\rvert)^x$. The value of $\lim\limits_{x\to\infty}\left[\frac{2}{f(x)}\right]$ equals (Here [ ] represents greatest integer function)	(S) 1

56.

Column A	Column B
(A) $\lim\limits_{x\to 1}\frac{\ln x}{x^4-1}$	(P) 2
(B) $\lim\limits_{x\to 0}\frac{3e^x-x^3-3x-3}{\tan^2 x}$	(Q) $\frac{2}{3}$
(C) $\lim\limits_{x\to\infty}\frac{\pi-2\tan^{-1}x}{\ln\left(1+\frac{1}{x}\right)}$	(R) $\frac{3}{2}$
(D) $\lim\limits_{x\to 0}\frac{2\sin x-\sin 2x}{x(\cos x-\cos 2x)}$	(S) $\frac{1}{4}$
(E) $\lim\limits_{x\to 0}\frac{e^x-e^{-x}-2x}{x-\sin x}$	

57.

Column A	Column B
(A) $\lim\limits_{x\to\infty}\left(\frac{x}{1+x}\right)^x$ equals	(P) e^2
(B) $\lim\limits_{x\to\infty}\left(\sin\frac{1}{x}+\cos\frac{1}{x}\right)^x$	(Q) $e^{-1/2}$
(C) $\lim\limits_{x\to 0}(\cos x)^{\cot^2 x}$	(R) e
(D) $\lim\limits_{x\to 0}\left(\tan\left(\frac{\pi}{4}+x\right)\right)^{1/x}$	(S) e^{-1}

58. If $\lim\limits_{x\to 0}\frac{\sin 3x+A\sin 2x+B\sin x}{x^5}=C$, then find the value of A+ B+C.

59. Find the value of $\lim\limits_{x\to 0}\frac{x^{6000}-(\sin x)^{6000}}{100\cdot x^2\cdot(\sin x)^{6000}}$

60. If $\lim\limits_{x\to 0}\frac{x+\ln\left(\sqrt{x^2+1}-x\right)}{x^3}=\frac{p}{q}$ where p and q are coprimes then find the value of $\lvert p^2-q^2\rvert$

61. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Find the limit of ratio of the areas of the two triangles thus formed as the arc AB decreases indefinitely

62. If $\lim\limits_{x\to 0}\frac{ae^x-b\cos x+ce^{-x}}{x.\sin x}=2$, then find the value of a + b + c.

63. Let $L=\prod\limits_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M=\prod\limits_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ and $\prod\limits_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$, then find the value of $L^{-1}+M^{-1}+N^{-1}$.

64. Find the value of

$$\lim_{x\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+........+\frac{1}{\sqrt{n^2+2n}}\right).$$

65. Find the value of $\lim\limits_{x\to 0^+} x^{\left(x^x-1\right)}$.

66. If $\lim\limits_{x\to 0} \dfrac{1-\cos 3x\cdot\cos 9x\cdot\cos 27x.......\cos 3^n x}{1-\cos\frac{1}{3}x\cdot\cos\frac{1}{9}x\cdot\cos\frac{1}{27}x.........\cos\frac{1}{3^n}x} = 3^{10}$, find the value of *n*.

67. If the $\lim\limits_{x\to 0}\dfrac{1}{x^3}\left(\dfrac{1}{\sqrt{1+x}}-\dfrac{1+ax}{1+bx}\right)$ exists and has the value equal to *l*, then find the value of $\dfrac{1}{8}\left(\dfrac{1}{a}-\dfrac{2}{1}+\dfrac{3}{b}\right)$

68. If the equations, $x^2+ax+12=0$, $x^2+bx+15=0$ & $x^2+(a+b)x+36=0$ have a common positive root, and the other three roots are also the roots of the polynomial $P(x)=x^3+ax^2+bx+c$, then evaluate $\lim\limits_{x\to 5}\dfrac{-P(x)}{(x-4)^5}$.

69. Evaluate $\lim\limits_{x\to\pi/3}\dfrac{3\tan x-\tan^3 x}{3\cos\left(x+\frac{\pi}{6}\right)}$

70. Evaluate

$$\lim_{x\to 1}\left[\left[\frac{4}{x^2-x^{-1}}-\frac{1-3x+x^2}{1-x^3}\right]^{-1}+\frac{3\cdot(x^4-1)}{x^3-x^{-1}}\right].$$

71. Evaluate $\lim\limits_{x\to 1}\dfrac{x(1-m\cos x)+n\sin x}{x^3}=1$, then find m + n.

72. Evaluate $\lim\limits_{x\to\infty}\left[\dfrac{x}{x+\dfrac{x^{1/3}}{x+\dfrac{x^{1/3}}{x+.....\infty \text{ terms}}}}\right]$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (2)
3. (1)
4. (4)
5. (3)
6. (3)
7. (3)
8. (4)
9. (2)
10. (2)
11. (4)
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13. (4)
14. (1)
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32. (1)
33. (4)
34. (2)
35. (2)
36. (3)
37. (4)
38. (2)
39. (1)
40. (1)
41 (1, 4)
42. (1, 2, 3, 4)
43. (2, 3)
44. (1, 3)
45. (1, 2, 4)
46. (1, 2, 4)
47. (1, 2, 3, 4)
48. (1, 2)
49. (2, 3)
50. (1, 2)
51. (1, 2, 3)
52. (3, 4)
53. (A) R; (B) S;(C) P;(D) Q
54. (A) Q; (B) R;(C) P;(D) P
55. (A) S; (B) P ;(C) Q; (D) R
56. (A) S; (B) R; (C) P; (D) Q ;(E) P
57. (A) S;(B) R;(C) Q; (D) P
58. (2)
59. (10)
60. (35)
61. (4)
62. (3)
63. (8)
64. (2)
65. (1)
66. (4)
67. (3/4)
68. (0)
69. (8)
70. (3)
71. (4)
72. (1)